

ASSINGNMENT 5 – REPORT

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Problem Overview

The objective was to build a binary classification model using anonymized tabular data to predict the target variable (0 or 1). This was framed as a supervised learning task.

Data Loading & Exploration

The training and test datasets were loaded from CSV files. Initial exploration included checking for missing values, data types, and the distribution of the target variable.

Feature Processing

- **ID Column Dropped:** It had no predictive value.
- **Mutual Information (MI)** was used for feature selection. MI measures how much information about the target is captured by each feature. Features with higher MI scores are more useful for prediction.
- The top MI-ranked features were selected for model training.
- **Scaling:** Selected features were standardized using `StandardScaler` to ensure fair contribution to model learning.

Model Training

Several models such as Logistic Regression, Decision Tree, and Random Forest were evaluated. Logistic Regression or a tree-based model was selected based on performance on the validation set.

Evaluation

Models were assessed using accuracy, precision, recall, and F1-score. These metrics helped compare models on how well they predicted each class.

Logistic Regression

Accuracy : 0.901625

Precision: 0.6430976430976431

Recall : 0.047512437810945274

F1-Score : 0.08848737549223998

Decision Tree

Accuracy : 0.896225

Precision: 0.3761814744801512

Recall : 0.04950248756218906

F1-Score : 0.08749175642998462

Random Forest

Accuracy : 0.900575

Precision: 0.8028169014084507

Recall : 0.014179104477611941

F1-Score : 0.02786604742116842

Kaggle Score & Submission

The final model was applied to the test data to generate predictions. These were saved in the submission format and uploaded to Kaggle.

So below are the Kaggle score with respective models with MI features :-

Models	Score
Logistic Regression	0.52941
Decision Tree	0.52278
Random Forest	0.50880